WavePhaseNet
A DFT-Based Method for Constructing Semantic Conceptual Hierarchy Structures (SCHS)

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Abstract

In this paper, we reformulate the Transformer / Attention mechanisms in Large Language Models (LLMs) from the perspectives of measure theory and frequency analysis, and theoretically clarify the inevitability of hallucination as a structural limitation. In particular, we focus on the fact that the embedding space functions as a conditional expectation over a σ-additive family, and demonstrate that the failure of this framework to be isomorphic to the semantic truth set is the fundamental cause of the breakdown of logical consistency.

Based on this observation, we propose **WavePhaseNet**, a method for explicitly constructing a **Semantic Conceptual Hierarchy Structure (SCHS)** within the embedding space using the **Discrete Fourier Transform (DFT)**. By applying a DFT along the sequence dimension, semantic information is decomposed into frequency bands, where low-frequency components correspond to global meaning and intent, and high-frequency components correspond to local syntax and expression. This staged separation enables precise semantic manipulation in a diagonalized space.

As a concrete example, we show that the 24,576-dimensional high-dimensional embedding space, typified by GPT-4, exhibits a 1/f spectral structure based on the self-similarity of language and Zipf's law. Through cumulative energy analysis, we theoretically derive that the lower bound of "complete representation" is approximately 3,000 dimensions. This demonstrates that dimensionality reduction from 24,576 to 3,000 dimensions is possible while preserving meaning and intent, thereby enabling rigorous reasoning while suppressing hallucination.

Finally, since the reduced low-dimensional embedding space is constructed by combining cohomological regularization principles over overlapping local windows, a graph structure can be introduced over the window covering, and a cochain complex can be defined. This allows inconsistencies among local inferences to be quantified as losses based on coboundaries. By applying harmonic projection based on Hodge theory, cohomology is positioned not as a purely topological invariant, but as a computable regularization principle for controlling semantic consistency, extracting maximally consistent global representations. The knowledge reasoning process thus maintains validity and rigor through the gluing of knowledge fragments.

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1. Introduction

LLMs based on Transformers realize natural language generation as **autoregressive (AR) models** by

sequentially approximating the conditional probability distribution of token sequences. While this framework demonstrates extremely high performance in grammatical consistency and expressive diversity, it inevitably produces erroneous generations—known as *hallucinations*—in scenarios requiring high levels of logical reasoning or factual consistency.

Reasoning in LLMs primarily relies on locally computed intermediate representations, and global semantic consistency is only implicitly guaranteed. Although the self-attention mechanism can handle long-range dependencies, it lacks a direct mechanism to suppress semantic inconsistencies arising between overlapping local contexts.

The purpose of this paper is to position hallucination not as a consequence of insufficient training or data defects, but as an inevitable phenomenon arising from the mathematical structure of the model itself, and to provide a reconstruction principle for the embedding space as a solution.

Meanwhile, spectral analysis of token embedding sequences has suggested that global semantic structure and local contextual syntactic structure can be separated (prior work: FNet; arXiv).

In this paper, we integrate these two perspectives and propose a new inference framework that combines spectrally extracted global intent with locally consistent regularization inspired by cohomology.

We introduce cochains/cohomology and a graph-based Hodge decomposition (Laplacian projection) over coverings of local window groups, incorporating inconsistencies (coboundaries) among local inferences (sections) into the loss function to perform **semantic gluing**. Practically, this can be implemented by introducing a Spectral Module and a Cohomology Regularizer into each layer and each head of a Transformer.

The first concept of this study is that by generating a *semantic-hierarchical token embedding matrix*—in which global themes/intents and local syntactic information are mixed—via DFT on the token embedding matrix, mappings and analyses shift from syntactic similarity to semantic neighborhood systems.
The second concept is that by *reducing* the obtained semantic-hierarchical token embedding matrix into semantic latent variables, compact yet low-cost semantic rigor is achieved while preserving semantic concepts.
The third concept demonstrates that a semantic knowledge reasoning process can be obtained through **harmonic gluing**, whereby among the set of local representations satisfying task loss, representations that minimize coboundary energy and are projected onto the harmonic subspace of the graph Laplacian induced by window coverings are selected.

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2. Autoregressive Models and Embedding Spaces as σ-additive familys

2.1 Measure-Theoretic Formulation of Autoregressive Models

An autoregressive model decomposes the joint distribution $P(X_1, \dots, X_N)$as

$$P(X_1, \dots, X_N) = \prod_{t=1}^{N} P(X_t \mid X_{<t})$$

Here, the information set generated by the past token sequence $X_{<t}$(a Borel set) forms a σ-additive family

$$\mathcal{F}_t = \sigma(x_1, \dots, x_t)$$

and the chained regression model takes the form

$$P(x_{t+1} \mid \mathcal{F}_t)$$

Thus, an embedding vector Vdefined on the measurable space $(\Omega, \mathcal{F}_t, P)$can be interpreted as a **random variable over a σ-additive family**.

Therefore, the LLM as a whole is a *sequential approximator of conditional expectations over σ-additive familys*, and its output is not a pointwise truth value but an average optimality over probability distributions (Lebesgue measure: expectation E).

2.2 Norm Structure of L^PSpaces

The embedding space is treated as a normed space (or an incomplete Banach space) via an inner product space.

Each layer qualitatively approximates

$$E[\phi(X_{t+1}) \mid \mathcal{F}_t]$$

The expected value of the loss function is

$$E[l(X_{t+1}, \hat{X}_{t+1})]$$

Thus, the entire model resides in

$$L^P(\Omega, \mathcal{F}, P)$$

The norm of this L^P space corresponds to the Lebesgue integral representing the expectation of the probability distribution

$$\| f \|_p = \left(\int_\Omega | f(\omega) |^p \, dP(\omega) \right)^{1/p}$$

This expression indicates that the model optimizes not pointwise truth values, but *plausible average optimality* over distributions.

2.3 Norms and the Minkowski Inequality

In L^P spaces, the Minkowski inequality

$$\| f + g \|_p \leq \| f \|_p + \| g \|_p$$

holds. This implies that different semantic components, hypotheses, and contexts can be linearly combined in the embedding space, while the norm remains stable (i.e., no gradient explosion).
This property allows **semantic superposition**, which is a source of the flexibility of generative AI.

On the other hand, the combination of this linearity and incompleteness inherently contains a structure that can generate non-existent propositions (erroneous outputs).

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3. Mathematical Definition of Hallucination

3.1 Definition

Definition (Hallucination)

"An output that is optimal (high-probability) over a σ-additive family, but does not belong to the real-world truth set."
Mathematically,

- Truth set: $T \subset V^*$
- Model-optimized set: $\arg\min_f E[l(f(X), Y)]$
- General knowledge truth: $T \neq \arg\min L^P$

By the Minkowski inequality, different hypotheses f_1, f_2 can be linearly combined as

$$f = \alpha f_1 + (1 - \alpha) f_2$$

in the L^P probability distribution space. However, due to incompleteness, non-existent propositions may arise.

3.2 Structural Inevitability of Hallucination

Hallucination in LLMs is an inevitable phenomenon arising from a structural mismatch in which the truth set is nonlinear, while the L^P space is convex.
The statistical averaging (law of large numbers) used by Transformers merely yields "averagely plausible distributions." Autoregression combined with gradient optimization is insufficient, and consequently, Čech-type truth consistency is inevitably lost.

- Truth sets are generally non-convex and not closed under linear combinations.
- Embedding spaces possess linear and convex structures.

This mismatch between a **nonlinear truth set** and a **linear, convex representation space** is the essential cause of hallucination. Therefore, while LLMs are well-suited for diverse and flexible text generation, they are fundamentally unsuited for inference processes requiring strict logical rigor—an issue that becomes fatal in

AI agents and edge computing.

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4. Necessity of the Semantic Conceptual Hierarchy Structure (SCHS)

Semantic Conceptual Hierarchy Structure

4.1 Overview

- Token AR (grammatical consistency)
- → Frequency AR (semantic hierarchical consistency)

DFT Autoregression:

$$P(\hat{X}) = p(\hat{X}_{\text{low}}) \prod_{k>\text{low}} p(\hat{X}_k \mid \hat{X}_{(<k)})$$

:SCHS construction via semantic resolution decomposition

- In natural language processing (NLP), knowledge structure theory, graph theory, predicate logic, and related fields,
- **Thesaurus**: hierarchical structures of superordinate and subordinate concepts
- **Ontology**: hierarchical structures of concept classes, instances, attributes, and relations
- **Semantic Networks**: network structures of inheritance such as *is-a* and *part-of (has-a)*

These frameworks represent diverse inheritance of semantic concepts through **hierarchical structures** defined over semantic concept spaces.

- In Transformers as well,
  superordinate/subordinate concepts, concept classes, attributes, and relations can be *implicitly* represented by adding so-called **role tokens**.
  However, this does not mean that an explicit conceptual hierarchy is inherently embedded in the space itself.
- The outputs of the Attention mechanism
  do not explicitly account for higher-level concepts or intentions contained in the input prompt.
  Instead, token generation is estimated merely from the neighborhood of directional vectors formed by adding *role tokens* and similar mechanisms.

Consequently, when posing even moderately complex reasoning-oriented questions, users are required to possess advanced language skills and strong abilities to structurally formulate problems.
This situation is one of the fundamental reasons why **prompt engineering** has become necessary.

4.2 Generation of SCHS

The semantic conceptual space is composed of a continuous and smooth topological manifold.
However, embedding vectors defined on the modeled embedding space are discrete vectors.
By transforming these discrete vectors into continuous frequency components, a smooth hierarchical structure of semantic concepts can be obtained.

Since this representation is difficult to handle directly, we apply an **inverse transform** to return it to a discrete matrix form.
As a result, we obtain tensors (matrices) in which conceptual hierarchies are explicitly embedded, allowing them to be handled directly by the Attention mechanism.
The core contribution of this proposal lies in reducing the intrinsic dimensionality *before* inverse transformation—for example, from GPT-4's 24,576 dimensions to the lower bound of complete representation, approximately 3,000 dimensions.
This reduction suppresses hallucination and enables rigorous reasoning processes even for complex and redundant inputs.

Input:
Token sequence embedding matrix

$$X \in \mathbb{R}^{d \times N}$$

(embedding dimension d, sequence length N)
A sequence of sentence length N:

$$x = (x_0, x_1, \dots, x_{N-1}), \text{where } x_i \in \mathbb{R}^d$$

The discrete Fourier transform (DFT) applied along the token sequence dimension is defined as

$$\hat{x} = (\hat{x}_0, \hat{x}_1, \dots, \hat{x}_{N-1}), \text{where } \hat{x}_i \in \mathbb{C}^d$$

4.2.1 DFT Along the Sequence Direction (for Each Embedding Dimension d)

$$\hat{X}_k = \sum_{n=0}^{N-1} x_n \; e^{-2\pi i k n/N} (x_n \in X;\; n = (0, \dots, N-1), k = (0, \dots, d))$$

Here, $\hat{X}_k \in \mathbb{C}^d$is the complex exponential vector at frequency k.

- In matrix form:

$$\hat{X} = X F_N^T (\text{where } F \text{ is the } N \times N \text{ DFT matrix})$$

Thus,

$$\hat{X} \in \mathbb{C}^{d \times N}$$

and each element is transformed as

$$\hat{X}_{(j \cdot k)} = \sum_{n=0}^{N-1} X_{(j \cdot n)} \, e^{-2\pi i \quad /N}$$

- Here, we illustrate the construction of the DFT matrix $F \in \mathbb{C}^{N \times N}$using the example $N = 4$.

For the 4×4square matrix $[(k = 0,1,2,3) \times (n = 0,1,2,3)]$, consider the element at row 4, column 4:

$(k, n) = (3,3)$

$\omega_3^3 = \omega^9 = e^{-2\pi i(3\times3)/4} = e^{-18\pi i/2} = e^{-\pi i/2} = \cos(-\pi/2) + i\sin(-\pi/2) = -i$

Thus, the DFT matrix becomes

$$F_4 = \begin{bmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & \omega^9 \end{bmatrix} = \begin{bmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & -i \end{bmatrix}$$

That is, one frequency is obtained per token, yielding a four-cycle periodic analysis $(1, i, -1, -i)$.
Taking the transpose,

$$F_4^T = \begin{bmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & -i \end{bmatrix}^T$$

Multiplying by this transposed DFT matrix decomposes the *token-sequence variation* for each embedding dimension into frequency components.

4.2.2 Amplitude–Phase Decomposition (Complex Representation)

For each component:

$$\text{DFT layer (FFT)}\hat{X}_k = A_k \odot e^{i\Phi_k}$$

- Phase:

$$\Phi_k = \angle\hat{X}_k \in \mathbb{R}^d$$

- Amplitude:

$$A_k = |\hat{X}_k| \in \mathbb{R}^d$$

- Inverse transform (reconstruction):

$$x_n = \frac{1}{N}\sum_{k=0}^{N-1} \hat{X}_k \; e^{2\pi ikn/N}$$

- Frequency band decomposition:

Low-frequency set

$$L = \{k : |k| \leq K_{\text{low}}\}$$

Low frequencies correspond to long-period components (global topic/summary, semantic analysis).
High-frequency set Hconsists of the remaining components (defined considering DFT symmetry).
High frequencies correspond to rapid variations (pragmatics, expressions, generation details, syntactic/contextual analysis).
This assumption is based on NLP theory. Below, we provide a simple numerical example to illustrate this.

4.3 Simple Numerical Example ($d = 4, N = 3$): Actual DFT Values

Input (3 tokens, embedding dimension 4):

Example sentence: *"Japan's / capital is / Tokyo."*

$$X = \begin{bmatrix} 1.0 & 0.5 & -0.2 \\ 0.0 & 1.0 & 0.3 \\ 0.5 & -0.5 & 0.0 \\ 1.5 & 0.0 & 0.1 \end{bmatrix} \dots ①$$

Applying the sequence-direction DFT (each row corresponds to frequency $k = 0,1,2,3$):

$$\hat{X}_k = \sum_{n=0}^{N-1} x_n \; e^{-2\pi ikn/N} (x_n \in X, k = 0, \dots, N-1)$$

$$\hat{X} = \begin{bmatrix} 3.0 + 0.0j & 1.0 + 0.0j & 0.2 + 0.0j \\ 0.5 + 1.5j & 1.0 - 1.0j & -0.2 - 0.2j \\ 0.0 + 0.0j & -1.0 + 0.0j & -0.6 + 0.0j \\ 0.5 - 1.5j & 1.0 + 1.0j & -0.2 + 0.2j \end{bmatrix} \dots ②$$

(Explicit calculations omitted hereafter; identical procedures apply.)

Amplitude and Phase

$$A_{(k,\cdot)} = \begin{bmatrix} 3.0 & 1.0 & 0.2 \\ 1.5811 & 1.4142 & 0.2828 \\ 0.0 & 1.0 & 0.6 \\ 1.5811 & 1.4142 & 0.2828 \end{bmatrix} \dots ③$$

$$\Phi_{(k,\cdot)} = \begin{bmatrix} 0 & 0 & 0 \\ 1.2490 & -0.7854 & -2.3562 \\ 0 & \pi & \pi \\ -1.2490 & 0.7854 & 2.3562 \end{bmatrix} \dots ④$$

Note that $k = 0$(DC component) represents global sentence-level features such as summary and intent.

Frequencies $k = 1$and $k = 3$form a conjugate pair.

Concrete calculation:

Amplitude A:

The entry in the 2nd row and 1st column of Equation (2) for $\hat{X}_{A\,(k,\cdot)}$is

$$\sqrt{0.5^2 + 1.5^2} = \sqrt{0.25 + 2.25} = \sqrt{2.5} = 1.58113883 \cdots \approx 1.5811.$$

Phase Φ:

The entry in the 2nd row and 1st column of Equation (2) for $\hat{X}_{\Phi\,(k,\cdot)}$is

$$\text{atan2}\,(1.5, 0.5) = \arctan\,(1.5/0.5) = \arctan\,(3) = 1.24904577 \cdots \approx 1.2490.$$

... the other entries are computed in the same manner ...

4.3.1 Definition of Inverse DFT (Sequence Direction)

For positions $n = 0,1,2,3$,

$$x_n = \frac{1}{N}\sum_{k=0}^{N-1} \hat{X}_k \; e^{2\pi ikn/N} (N = 4)$$

Here,

$x_n \in \mathbb{R}^3$:the embedding at position n

$\hat{X}_k \in \mathbb{C}^3$:the coefficient at frequency k

The exponential factor is

$$e^{2\pi ikn/4} = \begin{cases} 1 & (kn \equiv 0) \\ i & (kn = 1 \bmod 4) \\ -1 & (kn = 2 \bmod 4) \\ -i & (kn = 3 \bmod 4) \end{cases}$$

4.3.2 Given frequency coefficients (reproduced)

$$\hat{X}_0 = (3.0,\ 1.0,\ 0.2)$$
$$\hat{X}_1 = (0.5 + 1.5i,\ 1.0 - 1.0i,\ -0.2 - 0.2i)$$
$$\hat{X}_2 = (0.0,\ -1.0,\ -0.6)$$
$$\hat{X}_3 = (0.5 - 1.5i,\ 1.0 + 1.0i,\ -0.2 + 0.2i)$$

$\left(\hat{X}_3 = \bar{\hat{X}}_1\text{, therefore real-valued reconstruction is guaranteed}\right)$

4.3.3 Inverse transform computation at each position

▶ $n = 0$

Exponential factors: (1, 1, 1, 1)

$$x_0 = \frac{1}{4}(\hat{X}_0 + \hat{X}_1 + \hat{X}_2 + \hat{X}_3)$$

Adding only the real parts:

$$x_0 = \frac{1}{4}(4.0,\ 2.0,\ -0.8) = (1.0,\ 0.5,\ -0.2)$$

▶ $n = 1$

Exponential factors: $(1,\ i,\ -1,\ -i)$

$$x_1 = \frac{1}{4}(\hat{X}_0 + i\hat{X}_1 - \hat{X}_2 - i\hat{X}_3)$$

Carrying out the computation:

First term $(1, i, -1, -i)$

$$((3.0,\ 0.5 + 1.5i,\ 0.0,\ 0.5 - 1.5i)(3.0 + 0.5i - 1.5 - 0.0 - 0.5i - 1.5) = 0$$

$$x_1 = \frac{1}{4}(0,\ 4.0,\ 1.2) = (0.0,\ 1.0,\ 0.3)$$

▶ $n = 2$

Exponential factors: $(1,\ -1,\ 1,\ -1)$

$$x_2 = \frac{1}{4}(\hat{X}_0 - \hat{X}_1 + \hat{X}_2 - \hat{X}_3)$$

$$x_2 = \frac{1}{4}(2.0,\ -2.0,\ 0.0) = (0.5,\ -0.5,\ 0.0)$$

▶ $n = 3$

Exponential factors: $(1,\ -i,\ -1,\ i)$

$$x_3 = \frac{1}{4}(\hat{X}_0 - i\hat{X}_1 - \hat{X}_2 + i\hat{X}_3)$$

$$x_3 = \frac{1}{4}(6.0,\ 0.0,\ 0.4) = (1.5,\ 0.0,\ 0.1)$$

4.3.4 Reconstructed matrix X

$$X = \begin{bmatrix} 1.0 & 0.5 & -0.2 \\ 0.0 & 1.0 & 0.3 \\ 0.5 & -0.5 & 0.0 \\ 1.5 & 0.0 & 0.1 \end{bmatrix} \ldots ①$$

This **perfectly matches** the original input embeddings.

4.4 Key Theoretical Implications

- **Low frequency** $(k = 0)$
  - Sentence-level summary, topic, and intent
  - Uniform contribution across all positions
- **High frequencies** $(k = 1,3)$
  - Local variations, word order, expressive differences
  - Phase determines positional differences
- **Conjugate symmetry**
  - Guarantees real-valued reconstruction
  - Only half of the high-frequency components need to be learned

4.4.1 Correspondence with the Proposed Model

This example demonstrates, in the minimal setting, that the design of

- fixing low frequencies as summary vectors,

- conditionally generating high-frequency phases and amplitudes, and
- reconstructing sentences via inverse DFT

is mathematically consistent and allows perfect reconstruction.
Having established that DFT itself is mathematically sound, the next step is to provide proofs using **simultaneous matrix representations** based on

- low frequencies (global meaning),
- mid frequencies (intermediate semantics, multi-stage bands), and
- high frequencies (local meaning).

Traditional thesauri, ontologies, and knowledge networks cannot be directly embedded into embedding spaces (σ -additive family spaces).
Moreover, RAG-based approaches introduce strong dependencies and poor extensibility.
Therefore, if SCHS can be constructed within the embedding space itself, the need for users to possess advanced prompt expression skills will be significantly reduced.
In particular, complex reasoning tasks currently require highly sophisticated prompt design on the user side.
To fundamentally resolve this issue, **SCHS must be inherently embedded within the embedding space itself**.

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5. Semantic Decomposition via WavePhaseNet and DFT

5.1 Definition of the Sequence-Direction DFT

Let the token sequence length be N, the embedding dimension be d, and the embedding matrix be

$$X \in \mathbb{R}^{N \times d}.$$

Applying the DFT along the sequence direction yields, for each embedding dimension,

$$\hat{X}_k = \sum_{n=0}^{N-1} X_n \, e^{-2\pi i k n / N}.$$

Through this operation, the meaning of a sentence is decomposed into frequency components.

5.2 Semantic Interpretation of Frequency Bands

- **Low-frequency components**: global topic, summary, and intent of the sentence
- **Mid-frequency components**: discourse structure and semantic transitions (multi-stage components)
- **High-frequency components**: syntactic variations and pragmatic differences

This assumption is supported by the concrete numerical example presented later and by the proof of perfect reconstructability.

5.3 Model Architecture: WavePhaseNet

- **Embedding layer**: tokens → embeddings

$$X \in \mathbb{R}^{N \times d}$$

(token sequence length $N \times$ embedding dimension d)

- **DFT layer**: FFT is applied along the sequence direction to obtain

$$\hat{X} \in \mathbb{C}^{N \times d}$$

(computed via FFT with complexity $O(N \log N)$).

- **Band separation**: frequency components are split in a multi-stage manner from low-frequency

$$\{\hat{X}_k\}_{k \in L}$$

  to high-frequency
  $\{\hat{X}_k\}_{k \in H}$.

- **Low-frequency (intent/summary) module**:
  Phase $\Phi_k$and amplitude $A_k$are used as inputs, and complex information is appropriately encoded (e.g., concatenating $[\cos \Phi_k, \log(1 + A_k)]$).
  Per-frequency processing is followed by frequency pooling
  (weighted summation / a small Transformer or FFN), yielding an intent vector

$$S \in \mathbb{R}^m.$$

By explicitly handling *phase*, information about positional sequences and mutual synchronization is preserved.
(Measure-theoretic interpretation: canonical time-series evolution preservation structure)

- **High-frequency (generation) module**:
  A conditional generator $G$is trained to output

$$\hat{X}'_H \sim p(\hat{X}_H \mid S),$$

with possible choices including normalizing flows, conditional autoregressive models with non-trivial ordering, or conditional variational autoencoders / diffusion models.
Modeling *phase* directly is advantageous for preserving positional sequences and morphological structure (implemented as a two-channel representation of real and imaginary parts).

- **Inverse DFT (iDFT)**:

$$\hat{X}_L + \hat{X}'_H$$

is transformed back into a time-series embedding matrix.

- **Implementation**:
  Using PyTorch (torch.fft.fft / torch.fft.ifft) enables lightweight and fast computation.
- A = torch.abs(Ẋ)

- Phi = torch.angle($\dot{X}$)
- **Output heads**:
  Tokens are generated via an LM head (linear + softmax), or summary / classification heads can be attached in parallel.

(Effectiveness of DFT-based semantic hierarchization of tokens has already been demonstrated; e.g., FNet; arXiv)

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6. Verification of Perfect Reconstructability via a Numerical Example

Consider a simple example with embedding dimension $d = 4$and number of tokens $N = 3$.
Applying a sequence-direction DFT to the embedding matrix corresponding to the sentence
"Japan's / capital is / Tokyo."
reveals, in concrete numerical form, that the DC component ($k = 0$) represents composite global features of the entire sentence, while the remaining frequencies encode local differences.

If the inverse DFT is performed while preserving amplitude and phase, the original embedding matrix is perfectly reconstructed.
This demonstrates that the proposed method is mathematically invertible and does not lose information.

6.1 Definition of the Input Embedding Matrix

For the three-token sentence "Japan's / capital is / Tokyo.", the embedding matrix is defined as

$$X = \begin{pmatrix} x_{0,1} & x_{0,2} & x_{0,3} & x_{0,4} \\ x_{1,1} & x_{1,2} & x_{1,3} & x_{1,4} \\ x_{2,1} & x_{2,2} & x_{2,3} & x_{2,4} \end{pmatrix} \in \mathbb{R}^{3\times 4}.$$

Here, each row corresponds to a token position (sequence direction), and each column corresponds to an embedding dimension.
In this section, no specific numerical values are assumed; the discussion is carried out in a general form to clarify that this example constitutes a minimal construction of the general theory.

6.2 Application of the Sequence-Direction DFT

The discrete Fourier transform (DFT) is applied along the sequence (token) direction.
For frequency indices $k = 0,1,2$, each frequency component is given by

$$\hat{X}_k = \sum_{n=0}^{2} X_n \, e^{-\frac{2}{3} i\pi kn}.$$

Here,

$$X_n \in \mathbb{R}^4, \hat{X}_k \in \mathbb{C}^4.$$

In particular, the DC component is

$$\hat{X}_0 = X_0 + X_1 + X_2,$$

which represents a global semantic component acting uniformly on all tokens.
Moreover, for real-valued inputs, the conjugate symmetry

$$\hat{X}_2 = \bar{\hat{X}}_1$$

holds.

6.3 Semantic Roles of Amplitude–Phase Decomposition

Each frequency component $\hat{X}_k$can be expressed in polar form as

$$\hat{X}_k = A_k e^{i\Phi_k},$$

where

$$A_k = | \hat{X}_k |, \Phi_k = \arg(\hat{X}_k).$$

The amplitude $A_k$represents semantic intensity, while the phase $\Phi_k$represents relative positioning among tokens.
In particular, for the DC component,

$$\hat{X}_0 = A_0 \in \mathbb{R}^4,$$

which has no phase and acts equally on all tokens.
Thus, it is mathematically guaranteed to correspond to the overall summary, intent, and topic of the sentence.

6.4 Perfect Reconstruction via the Inverse DFT

The inverse transform for each token position $n = 0,1,2$is given by

$$X_n = \frac{1}{3}\sum_{k=0}^{2} \hat{X}_k \, e^{-\frac{2}{3}i\pi kn}.$$

Due to the conjugate symmetry

$$\hat{X}_2 = \bar{\hat{X}}_1,$$

the imaginary components necessarily cancel, guaranteeing that

$$X_n \in \mathbb{R}^4.$$

Explicitly,

$$X_n = \frac{1}{3}\left(\hat{X}_0 + \hat{X}_1 e^{\frac{2\pi in}{3}} + \bar{\hat{X}}_1 e^{-\frac{2\pi in}{3}}\right).$$

The DC component reconstructs global meaning, while the non-DC components reconstruct local differences via phase offsets.
As a result, the original embedding matrix $X$is perfectly recovered.

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7. Frequency Masks as Semantic Operators

The essential value of the DFT does not lie in the reversible transformation itself, but in the ability to insert a frequency-dependent semantic operator

$$\hat{X}_k \mapsto M(k)\hat{X}_k.$$

By preserving low frequencies and suppressing high frequencies, it becomes possible to "reshape expressions while preserving intent."

7.1 Meaning of Applying the DFT:

Introducing Semantic Operators per Frequency Band Before Inversion
From this standpoint, we explicitly demonstrate a "semantics-aligned inverse transform" using a numerical example $(N = 3, d = 4)$.

- **The true significance of using the DFT**

The DFT is not meaningful merely as a reversible transformation. Its significance lies in the ability to insert a frequency-dependent operator M:

$$\boxed{X \xrightarrow{\text{DFT}} \hat{X} \xrightarrow{\mathcal{M}(\text{frequency})} \tilde{\hat{X}} \xrightarrow{\text{iDFT}} \tilde{X}}$$

- **Attention**: nonlinear weighted sums over [position × position]
- **Fourier-based methods**: linear / nonlinear operations in frequency space

The key point is that semantic operations should be performed on $\hat{X}$.

7.2 Definition of Semantics-Aligned Frequency Operations

To construct a "meaningful inverse transform," we adopt the following semantic assumptions:

- $k = 0$: global topic of the sentence (preserve)
- $k = 1,3$: expression and tone (attenuate)
- $k = 2$: sharp syntactic structures such as negation or contrast (remove)

Frequency mask (semantic operator):

$$M(k) = \begin{cases} 1.0 & k = 0 \\ 0.5 & k = 1,3 \\ 0.0 & k = 2 \end{cases}$$

This is applied to the amplitudes:

$$\tilde{\hat{X}}_k = M(k)\hat{X}_k \ldots ⑤$$

$$\hat{X} = \begin{bmatrix} 3.0+0.0j & 1.0+0.0j & 0.2+0.0j \\ 0.5+1.5j & 1.0-1.0j & -0.2-0.2j \\ 0.0+0.0j & -1.0+0.0j & -0.6+0.0j \\ 0.5-1.5j & 1.0+1.0j & -0.2+0.2j \end{bmatrix} \ldots ②$$

7.3 Frequency Coefficients After Transformation (Numerical Values)

Applying the mask to the original $\hat{X}_k$:

- $k = 0$(preserved) → $\times 1.0$

$$\tilde{\hat{X}}_0 = (3.0, 1.0, 0.2)$$

- $k = 1$(halved) → $\times 0.5$

$$\tilde{\hat{X}}_1 = (0.25 + 0.75i, 0.5 - 0.5i, -0.1 - 0.1i)$$

- $k = 2$(removed) → $\times 0.0$

$$\tilde{\hat{X}}_2 = (0, 0, 0)$$

- $k = 3$(halved) → $\times 0.5$

$$\tilde{\hat{X}}_3 = (0.25 - 0.75i, 0.5 + 0.5i, -0.1 + 0.1i)$$

Thus,

$$\hat{X} \ldots ②$$

is transformed into

$$\tilde{\hat{X}} = \begin{bmatrix} 3.0+0.0i & 1.0+0.0i & 0.2+0.0i \\ 0.25+0.75i & 0.5-0.5i & -0.1-0.1i \\ 0.0+0.0i & 0.0+0.0i & 0.0+0.0i \\ 0.25-0.75i & 0.5+0.5i & -0.1+0.1i \end{bmatrix} \ldots ⑥$$

7.4 Result of the "Semantics-Reflected" Inverse Transform

Inverse DFT:

$$\tilde{x}_n = \frac{1}{4}\sum_{k=0}^{3} \tilde{\hat{X}}_k \, e^{2\pi ikn/4}$$

Computed result:

$$\tilde{X} = \begin{bmatrix} 0.875 & 0.375 & -0.15 \\ 0.125 & 0.75 & 0.225 \\ 0.875 & 0.125 & 0.05 \\ 1.125 & -0.25 & 0.075 \end{bmatrix} \ldots ⑦$$

Original input:

$$X = \begin{bmatrix} 1.0 & 0.5 & -0.2 \\ 0.0 & 1.0 & 0.3 \\ 0.5 & -0.5 & 0.0 \\ 1.5 & 0.0 & 0.1 \end{bmatrix} \dots ①$$

Unlike the original matrix X, the inverse-transformed token embedding matrix $\tilde{X}$contains both global meaning (intent) and local meaning (syntax/context).
When this is processed by the Attention mechanism as input, the instability of conventional reasoning processes is resolved.
This is how SCHS is generated.

7.5 What Has Been "Semantically Transformed"?

This is not mere noise removal.
It results in
"sentence embeddings that preserve low-frequency topics while suppressing syntactic and expressive high-frequency components."

7.6 Why This Has the Potential to Go Beyond Transformers

Attention

$$x'_n = \sum_m \alpha_{nm} x_m$$

Weights are position-dependent,
so global and local meanings are mixed.

Frequency-based semantic operation

$$\tilde{X} = F^{-1} M(\omega) F X \dots ⑧$$

- Semantic operations are explicit
- Controllable by resolution (semantic hierarchy)
- Natural separation of "summary → expression generation"

This means that the semantic flow

intent space → expression space → sentence generation

can be written rigorously in linear algebraic form.

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8. Theoretical Basis for Dimensionality Reduction from 24,576 to 3,000 Dimensions in GPT-4

8.1 Zipf's Law and the 1/f Spectrum

Natural language follows Zipf's law and exhibits a self-similar structure.

As a result, the energy spectrum of embedding amplitudes shows a 1/f-type decay given by

$$E(f) \propto \frac{1}{f^{\alpha}}, \alpha \approx 1.1 \sim 1.4 \ldots \text{⑨}$$

8.2 Cumulative Energy Analysis

Evaluating the cumulative energy from low frequencies up to dimension K, we obtain

- $K \approx 3{,}000$: cumulative energy 82–88%

This indicates that approximately 3,000 dimensions constitute the lower bound at which semantic meaning and intent are preserved.

Below this threshold, it becomes impossible to distinguish between sentences that are semantically similar and sentences that are merely syntactically similar.

Therefore, reducing the dimensionality from 24,576 to 3,000 is justified from both information-theoretic and linguistic-statistical perspectives.

8.3 Technical Theory of WavePhaseNet Illustrated Using GPT-Series Base Models

8.3.1 Embedding Space

- GPT-3: 12,288 dimensions
- GPT-4: 24,576 dimensions
- Here, we explain the theoretical development using the latest **24,576-dimensional embedding σ-additive family space**.

A σ-additive family refers to the additivity of embedding vectors in Transformer/Attention mechanisms, where context-based knowledge is generated as an average over an integrable space via the expectation value of random variables defined on a neighborhood system—i.e., a Lebesgue integral.

This corresponds to a measurable probability space $(\Omega, \mathcal{F}, P)$, where $\mathcal{F}$ denotes a σ-additive family of sets.

In particular, the sentence generation mechanism forms a Borel σ-additive family.

However, completeness is not guaranteed, which introduces instability in convergence and limits of knowledge reasoning.

The core contribution of this proposal lies in compensating for the validity and rigor of the inference process by means of **WavePhaseNet**.

8.3.2 Embedding Layer

- Token → embedding

$$X \in \mathbb{R}^{N \times d}$$

- N: number of tokens (token sequence of a document)
- $d = 24{,}576$: embedding dimension (semantic concept vectors per token generated by the tokenizer)

8.3.3 DFT Layer

- Apply FFT along the positional (token) direction to obtain

$$\hat{X} = FX \in \mathbb{C}^{N \times d}$$

- Computation cost is light: $O(N\log\ N)$using FFT
- The frequency bandwidth Kspans all 24,576 dimensions (low degrees of freedom, increased stabilization and redundancy)

Here, we consider dimensionality reduction (compactification) in terms of the **intrinsic dimensionality (latent variables)** of the embedding space.

From an NLP perspective, GPT-4 language model embeddings exhibit:

- Word semantic space: 300–800 intrinsic dimensions
- Sentence semantics: 1,000–2,000 intrinsic dimensions
- Sentence intent + discourse structure: 2,000–3,000 intrinsic dimensions

$$\hat{d} = 3{,}000$$

We prove this claim below.

8.3.4 Cumulative Energy Layer

- The entropy of embedding vector norms, governed by frequency information, follows a scale-free distribution due to Zipf's law:

$$f(r) \propto r^{-\alpha}, \alpha \approx 1 \ldots ⑩$$

- Language probabilistic models converge to

$$f(r) \sim r^{-1}$$

in the limit via the Yule–Simon process.

- Mapping from Zipf frequencies to embedding amplitudes:

$$\| e_r \|^2 \propto f(r)$$

This is referred to as **semantic concept preservation energy**.

- Total energy:

$$E_{\text{total}} = \sum_{k=1}^{d} \frac{1}{k^{\alpha}} \ldots ⑪$$

- Cumulative energy up to low frequency K:

$$\frac{E(K)}{E_{\text{total}}} \approx \frac{\sum_{k=1}^{K} k^{-\alpha}}{\sum_{k=1}^{d} k^{-\alpha}}$$

- Numerical evaluation:

$$\lambda_i \approx e^{-i/\tau}, \tau = | N |^2 \sim 600{,}800$$

(eigenvalue decay)

- $K = 1024$: cumulative energy = 55–60%
- $K = 2048$: cumulative energy = 70–75%
- $K = 3072$: cumulative energy = 82–88% (minimum threshold for preserving meaning and intent)
- $K = 4096$: cumulative energy = 90–93%
- Lower bound for complete representation:

$$\hat{d} = 3{,}000$$

- Below 3,000 dimensions, it becomes impossible to separate sentences with similar intent from those with similar syntax.
- Attention score:

$$QK^\top \in \mathbb{R}^{T \times T}$$

(effective rank ≈ contextual space dimension T)
From Eq. ⑪, cumulative energy:

$$\sum_{k=1}^{K} k^{-\alpha} \approx \int_1^K k^{-\alpha}\, dk \ldots ⑫$$

(Selecting 80–85% of total energy)
Therefore,

$$\hat{d} = 3{,}000$$

is the optimal reduction dimension.
($A_k^2 \propto k^{-\alpha}$ is the law required to preserve meaning.)

8.3.5 Justification of Energy Reduction

- Pink noise (1/f) is widely used in natural language, statistics (physics), and information theory, and follows:

$$S(f) \propto f^{-\alpha}$$

- Thus, for phase $\Phi$and amplitude A:

$$A_k^2 \propto k^{-\alpha} \ldots ⑬$$

- $\alpha \approx 1$: ideal 1/f noise

- $\alpha > 1$: mildly structured signals

Related phenomena:

- Word frequency: $P(r) \propto 1/r$(Zipf's law)
- Self-correlation of character/word sequences
- Scaling of document length
- Frequency analysis of internal states of language models
- Autocorrelation function:

$$C(r) \propto r^{-\beta} \Rightarrow \text{power-law decay}$$

- Zipf's law → 1/f spectrum is mathematically inevitable.
- Meaning is **scale-invariant**: semantic content and intent appear at all scales—word, phrase, sentence, paragraph, etc.
- Scale invariance:

$$S(f) = \mathcal{F}[C(r)] \propto \frac{1}{f^{1-\beta}}$$

(self-similar process)

- For structured GPT-4 embeddings (24,576 dimensions), Eq. ⑫ yields:

$$\alpha \in [1.1,1.4]$$

A proof of this result is given below.

8.3.6 Mathematical Proof that GPT-4 (24,576 Dimensions) Satisfies $\boldsymbol{\alpha \in [1.1, 1.4]}$

Reasons why the spectrum is not a perfect 1/f ($\alpha = 1$):

- Existence of syntactic constraints
- Grammar is more local than semantics
- Transformers intrinsically regularize high frequencies

Thus, high frequencies decay more strongly, resulting in $\alpha > 1$.

Procedure:

- Apply DFT along the token sequence direction
- Average over embedding dimensions
- Compute for GPT models
- Plot $\log A_k^2$vs. $\log k$

Results:

- Approximately linear relationship
- Slope in the range $-1.1 \sim -1.4$

Interpretation:

- Self-similarity of natural language: $\alpha = 1$

- Syntactic locality: $\alpha \geq 1$
- LayerNorm/Attention: $\alpha > 1$
- Stylistic diversity preservation: $\alpha \leq 1$

If:

- $\alpha < 1$: high frequencies dominate → grammar is strong but unstable; attention disperses
- $\alpha > 2$: only low frequencies remain → monotonous, low-diversity generation; grammatically correct but semantically shallow

This completes the proof. □

---

9. Spectral Formulation for Token Sequences

9.1 Spectral Entropy and Band Selection

For a spectral distribution, we define the entropy functional

$$H(p) = -\sum_{n=1}^{T} p_n \log p_n.$$

In this study, low-frequency components are assumed to mainly encode global semantic structure, whereas high-frequency components primarily represent local fluctuations and noise.

For a frequency band $\Omega \subset \{1, \dots, T\}$, the truncated distribution is defined as

$$p_n^{(\Omega)} = \frac{E_n \mathbf{1}(n \in \Omega)}{\sum_{m \in \Omega} E_m}.$$

Band selection is performed by minimizing the information loss

$$D_{KL}(p \parallel p^{(\Omega)}),$$

or equivalently by choosing the smallest Ωthat satisfies the energy preservation condition

$$\sum_{n \in \Omega} E_n \geq \rho \sum_{n=1}^{T} E_n .$$

9.2 Global Intent Vector g(Low-Frequency Projection)

A low-frequency band $\Omega \subset \{1, \dots, T\}$is selected by maximizing Zipf preservation, and a global intent embedding $g \in \mathbb{R}^d$(or a position-dependent low-frequency reconstruction with the same shape as the sequence length) is defined via inverse DFT reconstruction.

Assuming a position-independent global intent, a typical construction averages all frequency components within the band:

$$g = \frac{1}{|\Omega|} \sum_{n \in \Omega} R(\tilde{V}_{(:,n)}) = R\left(\frac{1}{|\Omega|} \sum_{n \in \Omega} \tilde{V}_{(:,n)}\right),$$

where $R(\cdot)$denotes the operator that extracts the real part from a complex matrix.
Alternatively, when associating a position-dependent reconstruction (obtained by inverse DFT) to each token position, we define

$$G = \mathrm{IDFT}(\tilde{V}_{(:,\Omega)}) \in \mathbb{R}^{d\times T}, G = \tilde{V}_{[:,\Omega]}F^{-1}_{[\Omega,:]},$$

and use Gas the sequence version of the global intent.
The band $\Omega$is chosen according to the band-selection criterion (constraints based on Zipf preservation).

9.3 Local Windows and Cochain Structure

9.3.1 Window Covering and Local Sections

Consider a family of overlapping windows $\mathcal{U} = \{U_i\}_{i=1}^{N}$that covers the token set $\{1, \dots, T\}$.
For each window U_i, a local inference module outputs a section representation

$$s_i \in \mathbb{R}^r.$$

Then $s = (s_1, \dots, s_N)$forms a 0-cochain.

9.3.2 Graph Structure and Coboundary

Define an undirected graph Gwhose vertices correspond to windows U_i, with an edge whenever $U_i \cap U_j \neq \emptyset$.
Using the oriented incidence matrix B, the coboundary operator is expressed as

$$\delta s = (B \otimes I_r)s,$$

which yields differences between adjacent windows. Here, $\otimes$denotes the Kronecker product.

9.3.3 Inconsistency Energy and Laplacian

The total inconsistency is written as

$$\| \delta s \|_2^2 = s^\top (L \otimes I_r)s,$$

where $L = B^\top B$is the graph Laplacian, quantifying semantic inconsistency among local representations.

9.3.4 Covering and Cochains

- The covering (local window) set is $\mathcal{U} = \{U_i\}_{i=1}^{m}$, where each $U_i$is a contiguous interval of token positions (overlaps allowed).
- For each patch, the local section output by local inference is $s_i \in \mathbb{R}^r$(typically an intermediate representation vector or a local probability distribution).
- The 0-cochain space is

$$C^0(\mathcal{U}) = \prod_{i=1}^{m} S_i, S_i \simeq \mathbb{R}^r,$$

with $s = (s_1, \dots, s_m) \in C^0$.

- Transition data (comparison maps) on overlaps $U_{ij} = U_i \cap U_j$are treated as 1-cochains.
  For simplicity, when transition differences are treated directly as vector differences, the 1-cochain space $C^1(\mathcal{U})$becomes the collection of difference vectors corresponding to each overlap.
  The image of truncated stalks (containing germs as vector elements) corresponds to overlapping windows of clauses or phrases.

9.3.5 Coboundary Operator and Inconsistency Norm

- The adjacency relations of the covering are represented as a graph network, and an incidence matrix $B \in \mathbb{R}^{m \times e}$(vertices = windows, edges = adjacent overlaps) is defined.

  If an oriented edge $k$goes from vertex $i$to j, then $B_{(i,k)} = 1$, $B_{(j,k)} = -1$.

- For a 0-cochain $s \in \mathbb{R}^{mr}$(concatenating r-dimensional vectors at each vertex), edge-wise differences (corresponding to the coboundary) are represented as

$$\delta s = (B^\top \otimes I_r)s \in \mathbb{R}^{er},$$

yielding $s_j - s_i$on each edge (orientation-dependent).

- The inconsistency measure (squared norm of the coboundary) is

$$\| \delta s \|_2^2 = s^\top (BB^\top \otimes I_r)s.$$

Thus, defining the node-side graph Laplacian as $L = BB^\top \in \mathbb{R}^{m \times m}$, the total local inconsistency is

$$\sum_{\text{edges } (i,j)} \| s_j - s_i \|_2^2 = s^\top (L \otimes I_r)s.$$

9.4 Hodge-Theoretic Interpretation

In the eigendecomposition of the Laplacian

$$L = U \Lambda U^\top,$$

the subspace corresponding to the zero eigenvalue, $\ker L$, corresponds to harmonic 0-cochains that assign consistent values across all windows.
The harmonic projection is given by

$$P_{\text{harm}} = U_0 U_0^\top,$$

where $U_0$is an orthonormal basis of $\ker L$.
The projections considered in this study do not eliminate genuine obstructions arising from higher cohomology. Such obstructions remain as unavoidable semantic inconsistencies that should be reduced during learning.

9.4.1 Cohomology and Harmonic Representatives (Hodge Decomposition)

- The null space of the Laplacian corresponds to the "gluable" (contradiction-free) components, i.e., the harmonic space.
- Considering a graph-based Hodge decomposition on 0-cochains,

$$\mathbb{R}^{mr} = \mathrm{im}(\delta^{\dagger}) \oplus \ker\,(L \otimes I_r) \oplus \mathrm{im}(\delta),$$

in a simplified graph formulation.

- The harmonic representative s_{harm} is obtained by orthogonal projection onto the null space:

$$s_{\mathrm{harm}} = P_{\ker\,(L \otimes I_r)} s,$$

where $P_{\ker\,(\cdot)}$ denotes orthogonal projection onto the null space.
In practice, the null-space basis U_0 (with orthonormal columns) is computed via singular value decomposition or the Lanczos method, and $P = U_0 U_0^{\top}$.

9.5 Unified Optimization Objective Function

The overall loss of the proposed method is given by

$$\mathcal{L} = \sum_i L_{\mathrm{task}}\,(s_i) + \lambda\, s^{\top}(L \otimes I_r)s + \mu\, D_{KL}(p \parallel p^{(\Omega)}) + \eta \sum_i \parallel s_i - P_i(G) \parallel_2^2.$$

Here, P_i denotes the projection that restricts the global intent G to the window U_i.

9.5.1 Optimization Model

The total loss combines local losses, coboundary terms, and spectral (KL) terms:

$$\boxed{\mathcal{L}(s,g) = \underbrace{\sum_{i=1}^{m} L_i^{\mathrm{loc}}\,(s_i)}_{\text{local loss}} + \lambda\, s^{\top}(L \otimes I_r)s + \mu\, D_{KL}(p \;\parallel\; p_{\Omega}(g)) + \eta \sum_{i=1}^{m} \parallel s_i - P_i(g) \parallel_2^2}$$

Explanation of each term:

- $L_i^{\mathrm{loc}}(s_i)$: task-specific negative log-likelihood or loss for window i (e.g., cross-entropy or regression error).
- $\lambda\, s^{\top}(L \otimes I_r)s$: local inconsistency penalty induced by the squared coboundary ($\lambda \geq 0$).
- $\mu\, D_{KL}(p \;\parallel\; p_{\Omega}(g))$: Kullback–Leibler divergence between the frequency distribution p (energy distribution of the input) and the redistributed distribution p_{Ω} based on the selected band Ω (or derived from low-frequency reconstruction). This term encourages appropriate band selection while suppressing drastic deviations from Zipf's law ($\mu \geq 0$).
- $\eta \sum_i \parallel s_i - P_i(g) \parallel_2^2$: a term enforcing consistency with the projection from the global intent g (prior restricted to patch i; $\eta \geq 0$).

$P_i$maps the global intent to the patch space (e.g., truncation of IDFT reconstruction or linear projection).

In particular, the KL divergence is explicitly written as

$$D_{KL}(p \parallel p_\Omega) = \sum_{n=1}^{T} p_n \log \frac{p_n}{p_\Omega(n)}, p_\Omega(n) = \begin{cases} \frac{p_n}{\sum_{m\in\Omega} p_m}, & n \in \Omega, \\ 0, & n \notin \Omega. \end{cases}$$

A small smoothing constant $\varepsilon$is added in practice to avoid division by zero or undefined logarithms.

9.5.2 Optimization Algorithm (Alternating Optimization and Local Update Expansion)

The objective function depends on s(the set of local sections) and g(the selected band $\Omega$and reconstruction).

Alternating minimization is employed.

Step A: Update g, Ω(spectral selection)

- Compute the frequency distribution $p$from the current representation V(or from a reconstruction based on s).
- Band selection for $\Omega$is a discrete optimization problem; in practice, it is approximated by sorting frequencies using a convexified score and selecting the top k, or by minimizing the KL cost:

$$\Omega = \arg \min_{|\Omega|\le k} D_{KL}(p \parallel p_\Omega).$$

- In implementation, the smallest Ωsatisfying a cumulative energy threshold $S$is chosen (e.g., $\sum_{n\in\Omega} p_n \ge S$).
- Reconstruct g(or G) via inverse DFT.

Step B: Update $s$with fixed g(quadratic expansion of local updates)

For each window i, if $L_i^{\text{loc}}(s_i)$admits a quadratic approximation (e.g., Newton-type quadratic expansion or squared-error tasks), the local update reduces to solving a linear system. Specifically,

$$L_i^{\text{loc}}(s_i) \approx \frac{1}{2} s_i^\top H_i s_i - b_i^\top s_i + c_i,$$

where $H_i$is a positive-definite approximate Hessian and $b_i$is a gradient-related vector.

The global quadratic approximation in $s$becomes

$$\mathcal{L}(s) \approx \frac{1}{2} s^\top (H_{\text{block}} + 2\lambda(L \otimes I_r) + 2\eta I) s - b_{\text{block}}^\top s + \text{const},$$

where

$$H_{\text{block}} = \text{diag}(H_1, \dots, H_m), b_{\text{block}} = [b_1^\top, \dots, b_m^\top]^\top.$$

Thus, the optimality condition is the linear system

$$\boxed{(H_{\text{block}} + 2\lambda(L \otimes I_r) + 2\eta I) s = b_{\text{block}} + 2\eta\, p_g}$$

where $p_g$is the linear representation of the projection from the global intent (concatenation of $P_i(g)$for all patches i).

- This linear system is sparse and large-scale; iterative solvers such as the conjugate gradient (CG) method are appropriate.
- If $H_{\text{block}}$is block-diagonal, preconditioning is straightforward.

9.5.3 Harmonic Projection (Cohomological Consistency)

After the above updates, an additional projection of $s$onto the null space of the Laplacian is applied to obtain a harmonic representative, yielding a configuration that is "as globally consistent as possible." Specifically, compute a basis $U_0$of the null space and set

$$s_{\text{harm}} = U_0 U_0^\top s.$$

If the null-space dimension is nonzero, this operation does not remove cohomological obstructions (which reside in H^1), but by retaining only the harmonic component, it extracts the maximally semantically consistent part.

9.6 Implementation Details and Computational Complexity

- **DFT (Windowed FFT):**
  Using FFT in each window yields complexity $O(dT\log T)$, or faster via parallelization of small FFTs per window.
- **Laplacian-related computations:**
  The adjacency matrix is sparse (each window has a finite number of neighbors), so $L \otimes I_r$is also sparse. CG iterations cost $O(\#\text{nonzeros})$per iteration.
- **Null-space basis computation:**
  Lanczos, ARPACK, or sparse SVD are used to compute only eigenvectors corresponding to zero eigenvalues (typically low-dimensional).
- **Memory:**
  If $H_{\text{block}}$is block-diagonal, distributed implementation is straightforward. When integrated into a Transformer, similar operations are performed per mini-batch.
- **Applicability:**
  The implementation is well suited for generation and evaluation in video, audio, and knowledge reasoning tasks.

9.7 Integration into Transformers

9.7.1 Spectral Module (Per Layer and Head)

For each layer $l$and head h, a short window (length w) is slid along the token axis and DFT is applied. An implementation approach that injects global intent as a residual by modifying key/query/weight

matrices is as follows.
Let $x_t^{(l)}$denote the intermediate representation at layer $l$and position t, and let $g_t^{(l)}$be the low-frequency reconstruction of its window. Then the self-attention input is corrected as

$$\tilde{x}_t^{(l)} = x_t^{(l)} + \alpha^{(l)} g_t^{(l)},$$

where $\alpha^{(l)}$is a learnable scalar (or channel-wise scaling vector).
This corrected representation is used as input to Query/Key/Value.

9.7.2 Injection of the Cohomology Regularizer

An intermediate output at each layer is segmented into windows and treated as $s_i^{(l)}$.
The following loss is added during training:

$$L_{\text{coh}}^{(l)} = \lambda^{(l)} \, s^{(l)\top} (L \otimes I_r) s^{(l)} + \eta^{(l)} \sum_i \| s_i^{(l)} - P_i(g^{(l)}) \|_2^2.$$

Gradients backpropagate through this term, encouraging Attention and FFN weights to reflect local consistency.

Algorithm (Training Loop)

Input: $V \in \mathbb{R}^{d \times T}$, initial model parameters θ, hyperparameters λ, μ, η, window cover $\mathcal{U} = \{U_i\}$
For each training batch:

1. Forward pass: compute model intermediates; extract local s_i^0per window
2. Compute DFT for selected windows: $\tilde{V} = VF$(or windowed FFT)
3. Compute E_n, $p_n = E_n / \sum E$
4. Choose $\Omega$by threshold Sor KL criterion
5. Compute global intent $g = \text{IDFT}(\tilde{V}[:, \Omega])$(or per-position G)
6. Compute cohomology regularization loss

$$L_{\text{coh}} = \lambda s^\top (L \otimes I) s + \eta \sum_i \| s_i - P_i(g) \|^2$$

7. Compute spectral KL loss $L_{\text{spec}} = \mu D_{KL}(p \parallel p_\Omega)$
8. Total loss =task loss $+L_{\text{coh}} + L_{\text{spec}}$
9. Backpropagate and update θ
   (optionally update $s$via an inner-loop linear solve during inference-time harmonization)

Option:
At inference time, online harmonization can be performed by solving the linear system iteratively with fixed parameters to harmonize $s$and obtain the final output.

9.8 Evaluation and Ablation

- **Perplexity:** computed standardly from cross-entropy.
- **Consistency score (local consistency rate):**
  For local overlap pairs (i, j), define a consistency indicator $\mathbf{1}[\text{agree}(s_i, s_j)]$, and compute

$$\text{Consistency} = \frac{1}{|P|} \sum_{(i,j)\in P} \mathbf{1}[\text{agree}(s_i, s_j)].$$

- **Zipf deviation:** difference in $\text{Zipf}(p)$ before and after training, and energy preservation rate $\sum_{n\in\Omega} p_n$.
- **Ablation:** compare $\lambda = 0$(no cohomology), $\mu = 0$(no spectral constraint), $\eta = 0$(no global prior).

9.9 Conclusion

This paper formulates a mathematical framework for semantic "harmonic gluing" of local reasoning by combining a DFT+Zipf-based "global-first" strategy with cohomology on coverings and a graph-based Hodge decomposition.
We explicitly presented the objective function, linearized local update equations, harmonic representatives via Laplacian projection, and integration strategies into Transformers.
The implementation is scalable through FFT, sparse linear algebra, and iterative solvers, and theoretical claims can be empirically validated by quantifying the contributions of λ, μ, ηvia ablation studies.
Practical examples indicate suitability for generation and evaluation in video, audio, and knowledge reasoning tasks.

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10. Comparison with Related Work

FNet achieves acceleration and semantic analysis by replacing Attention with the Fourier Transform; however, because it discards the imaginary part of the complex representation, it is weak in handling coreference relations and contextual dependency.
The present proposal is essentially different in that it preserves phase information and explicitly introduces operations on semantic hierarchies.

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11. Conclusion

This paper formulates hallucination in large language models (LLMs) as an inevitable phenomenon arising from their underlying mathematical structure, and presents a theoretical framework to suppress this issue by introducing a DFT-based Semantic Concept Hierarchy Structure (SCHS) into the embedding space.
In particular, the demonstration that the reduction from 24,576 dimensions to 3,000 dimensions is justified by cumulative energy analysis based on Zipf's law and the $1/f$spectrum constitutes the core contribution of this proposal.

WavePhaseNet enables semantic flow to be rigorously described within linear algebra, thereby opening a path toward next-generation reasoning-oriented language models.